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Department of Information Technology

III B.Tech I Sem - AD

Industry Oriented Internship – Deep Learning Techniques Development

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Financial Document Tampering Detection System Using EfficientNetV2B0 and Grad-CAM

Batch no. 13

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Guide

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Introduction:

Financial document fraud has become increasingly common with the widespread use of digital editing tools. Documents such as invoices, receipts, bank statements, and bills can be manipulated to alter financial information, resulting in significant losses for organizations. Traditional manual verification methods are time-consuming and prone to human error. This project proposes an Explainable AI-based system that automatically detects whether a financial document is genuine or tampered. The system uses EfficientNetV2B0 for document classification and Grad-CAM to highlight suspicious regions, making the prediction transparent and easy to understand.

Literature Survey:

Several researchers have applied deep learning techniques to document forgery detection using Convolutional Neural Networks (CNNs), EfficientNet, and Vision Transformers. These methods have shown improved accuracy compared to traditional image processing approaches. Explainable AI techniques such as Grad-CAM have also been introduced to visualize the regions responsible for model predictions. However, many existing systems focus on document classification without providing clear visual explanations, making their decisions difficult to interpret.

Limitations of Existing Algorithms/Methodologies:

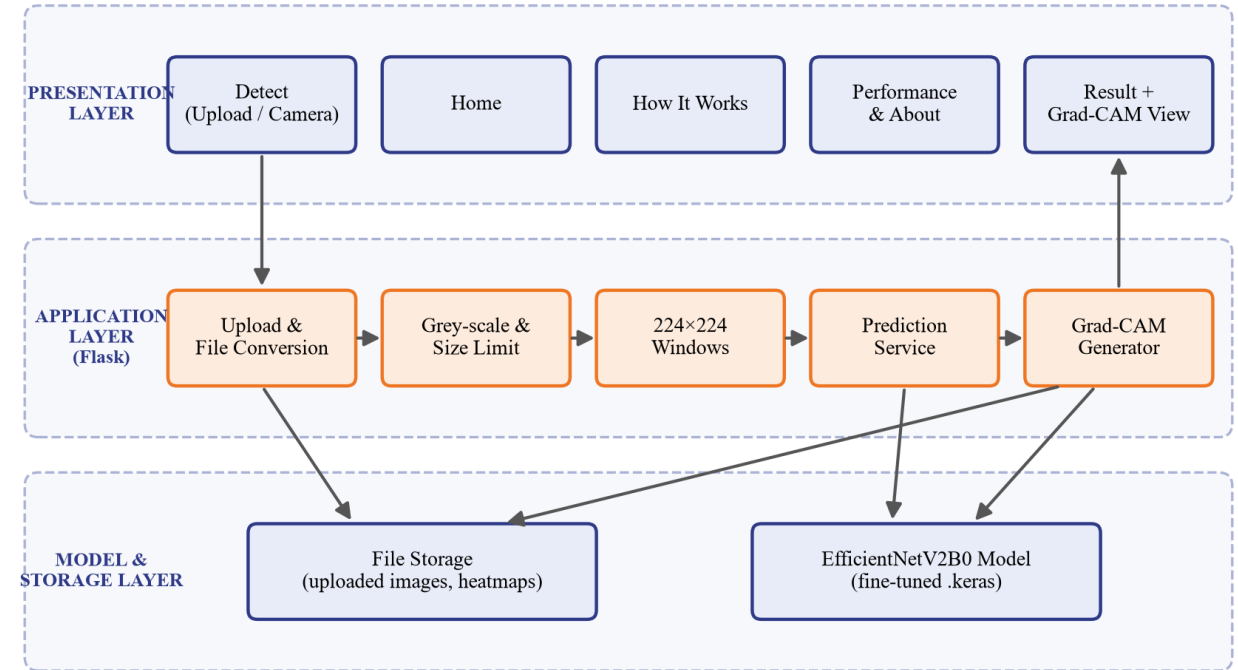
- Manual verification is slow and depends on human expertise.
- Traditional image processing techniques cannot detect complex digital manipulations.
- Many deep learning models act as black boxes and do not explain their predictions.
- Existing systems often fail to generalize across different types of financial documents.
- Several methods require expensive computational resources or specialized datasets.

Abstract:

Financial document tampering is a serious problem for banks, insurance companies and businesses. This project presents FinDoc AI, an explainable tampering detection system based on EfficientNetV2B0 and Grad-CAM, trained on the public “Find it again!” dataset of 988 scanned receipts. The model is fine-tuned on 224×224 windows cut at the receipts’ original resolution, and whole documents are scanned with overlapping windows; Grad-CAM heatmaps and a plain-language list of reasons explain every decision. The Flask web application accepts images, camera photos, PDF, Word and PowerPoint files, and checks the metadata of digital files. An earlier model that reported 97.85% patch accuracy was found to perform at chance level on whole receipts; the final system reaches 43.58% accuracy and a document-level AUC of 47.04% on 218 test receipts, so it is best used as an explainable screening aid.

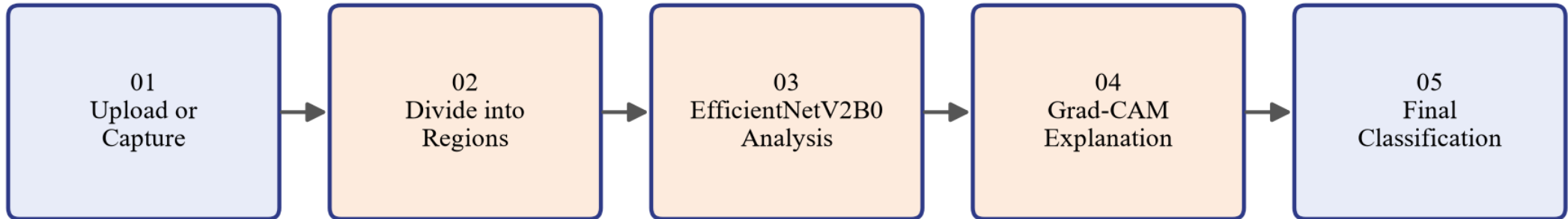
Proposed System:

- Upload an image, PDF, Word or PowerPoint file, or capture with the camera
- Scanned pages: grey-scale, 224×224 windows every 112 px, EfficientNetV2B0 score per window
- Document score = most suspicious window, compared with a threshold chosen on validation data
- Grad-CAM heatmap, flagged regions and plain-language reasons
- Digital files: metadata check (editor, dates, re-saves) instead of pixels



Methodology:

Training patches are 224×224 windows cut at the original resolution: 1,638 tampered windows around the 273 annotated edits and 2,695 genuine windows. EfficientNetV2B0 (ImageNet weights) is fine-tuned in two phases with balanced class weights.



Regional scores are combined into a document-level decision of Genuine or Tampered

Hardware Requirements:

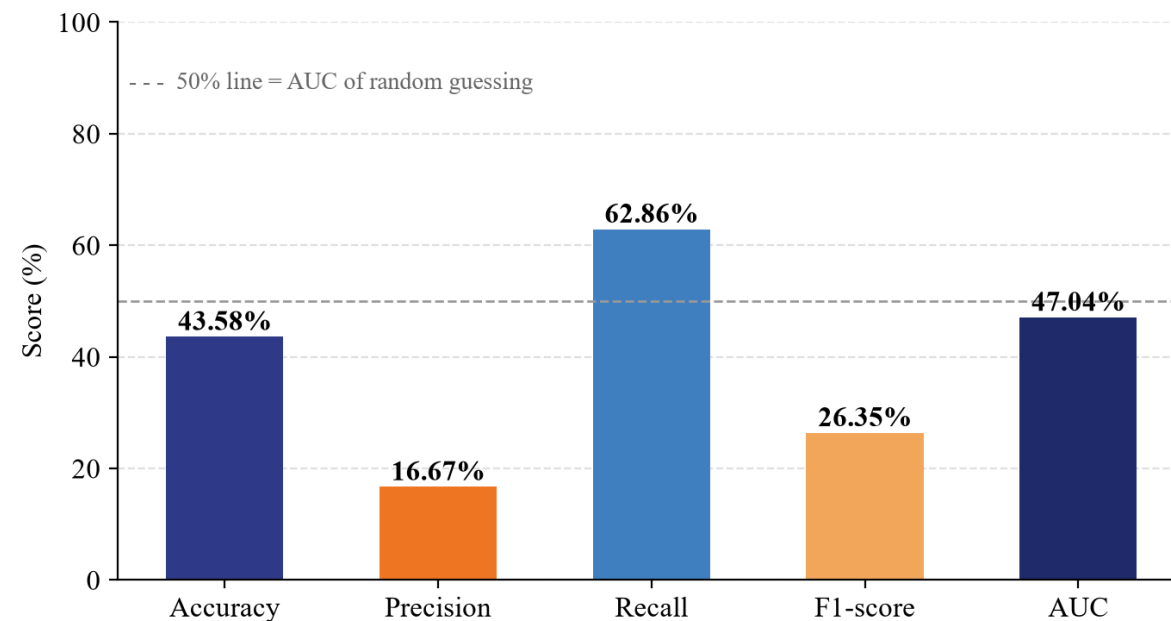
- Processor: Intel Core i5 or above
- RAM: 8 GB (16 GB for training)
- Storage: 10 GB free space
- Camera (optional, for capturing documents)

Software Requirements:

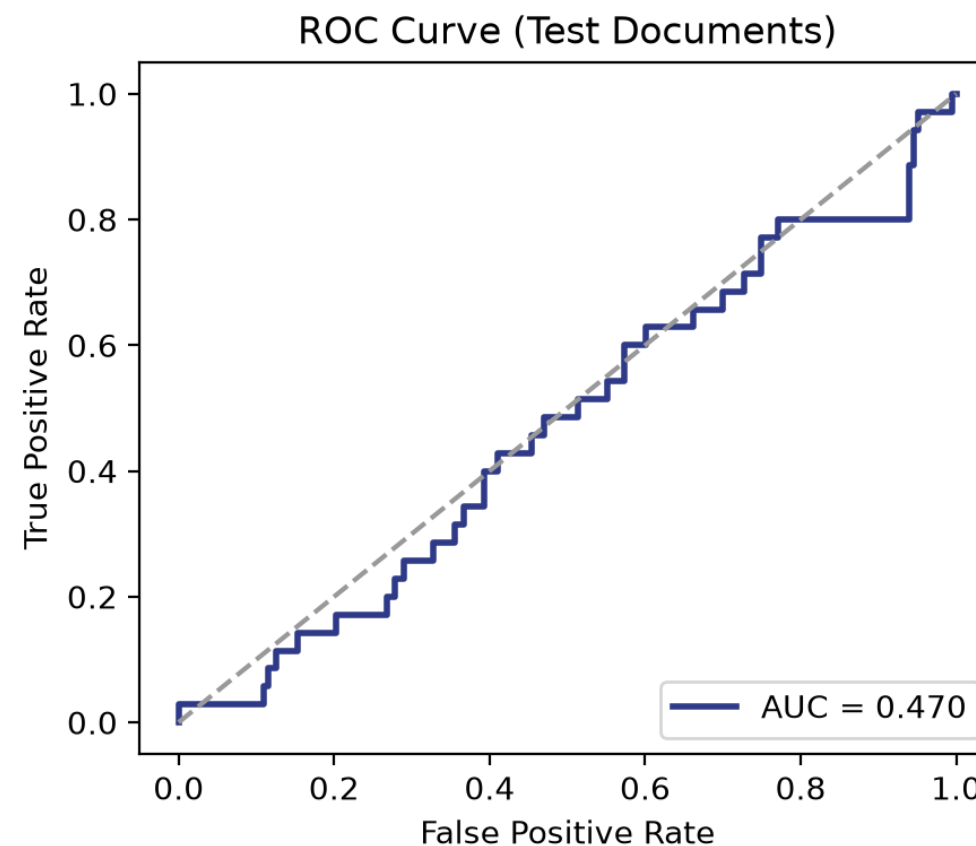
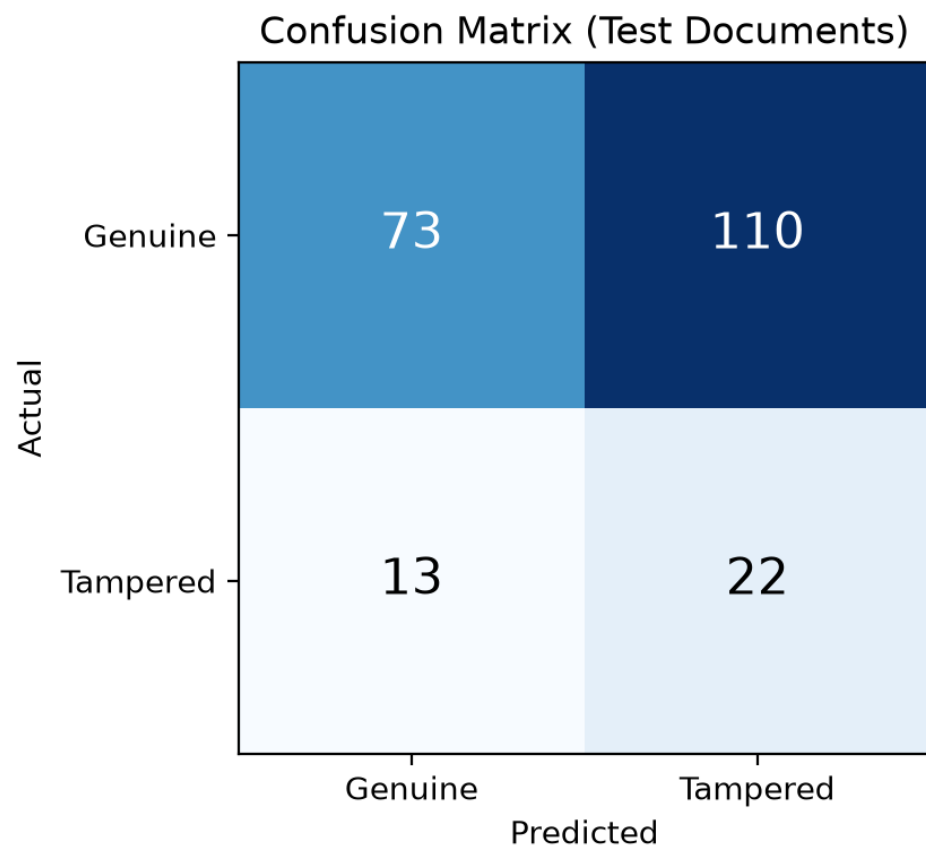
- Windows 10/11, Python 3.10+ (tested with 3.13)
- TensorFlow 2.20 with Keras 3.12 (EfficientNetV2B0)
- Flask 3.1, HTML, CSS, JavaScript
- Pillow, NumPy, PyMuPDF, scikit-learn, matplotlib
- Microsoft Word / PowerPoint (for DOC, DOCX, PPT, PPTX files)

Results on 218 Test Receipts:

- Accuracy: 43.58%
- Precision: 16.67%
- Recall: 62.86%
- F1-score: 26.35%
- ROC-AUC: 47.04% (50% = random guessing)
- Tampered receipts caught: 22 of 35
- Genuine wrongly flagged: 110 of 183



Confusion Matrix and ROC Curve:



Why the Earlier 97.85% Was Misleading:

- Old tampered patches: tiny crops (~ 80 px) enlarged $3\times$; genuine patches: 256 px crops shrunk
- The model learned the zoom level, not the tampering
- Tested on whole receipts: 49.5% accuracy, AUC 0.44 (chance level)
- Fix: every patch is a 224×224 window at the original scale, and evaluation uses whole documents
- Final model: detects 22 of 35 tampered receipts (was 14) but flags 110 of 183 genuine ones (was 89)
- The final numbers are lower, but they describe what users of the website actually get

Screenshots – Tampered Receipt Detected:

FD

FinDocAI

Home

Detect

How It Works

Performance

About

● ANALYSIS COMPLETE

Document Analysis Result

Your financial document has been analyzed using EfficientNetV2B0 and Grad-CAM.

FINAL RESULT

! Tampered

Tampering score
91.68%

Suspicious visual patterns were detected in one or more document regions.

WHY THIS RESULT

Reasons for the decision

File: X51008099081.png

- The most suspicious region scored 91.7%, which is at or above the decision threshold of 77.8%, so the document is reported as tampered.
- 7 of the 105 regions analysed scored above the threshold (several places look inconsistent).
- The strongest suspicion is in the bottom-right part of the page (outlined on the Grad-CAM map). Red and yellow areas in the heatmap are the pixels that pushed the tampering score up the most.
- Reliability: on 218 test documents the model reached 43.58% accuracy and caught 22 of 35 tampered documents, so treat this result as a screening signal, not proof.

REGIONS ANALYZED

105

Document regions processed

HIGHEST AI SCORE

91.68%

Most suspicious region

AVERAGE AI SCORE

43.52%

Average regional score

FLAGGED REGIONS

7

Regions above the detection threshold

STEP 02

AI Attention Map

Full-document Grad-CAM visualization

ORIGINAL DOCUMENT

GRAD-CAM EXPLANATION

HOW TO READ THE MAP

● Higher AI attention

● Moderate attention

● Lower attention

Screenshots – Digital Files (PDF, Word, PowerPoint):

- Typed Word, PowerPoint and digital PDF files have no scan traces
- The system checks the file metadata instead
- Example: an invoice PDF whose amount was changed and re-saved with iLovePDF
- Result: 3 editing signals found

The screenshot shows the FinDocAI website interface. At the top, there's a navigation bar with links: Home, Detect, How It Works, Performance, and About. The main header area displays 'FD FinDocAI' and 'ANALYSIS COMPLETE'. The central section is titled 'Document Analysis Result' with a subtitle: 'Your financial document has been analyzed using EfficientNetV2B0 and Grad-CAM.' Below this, a yellow box highlights 'Signs of editing found' with a sub-header 'DIGITAL DOCUMENT · DIGITAL PDF'. It includes an icon 'i' and states 'The file's metadata shows that it was changed after it was created.' To the right, it says 'Editing signals 3'. A section titled 'WHY THIS RESULT' contains 'Reasons for the decision' and lists six bullet points explaining the analysis. At the bottom, there's a 'PAGE 1 (RENDERED)' preview of an invoice and a 'HOW IT WAS DECIDED' section titled 'Analysis steps' with two numbered points.

FD FinDocAI

Home Detect How It Works Performance About

• ANALYSIS COMPLETE

Document Analysis Result

Your financial document has been analyzed using EfficientNetV2B0 and Grad-CAM.

DIGITAL DOCUMENT · DIGITAL PDF

Signs of editing found

The file's metadata shows that it was changed after it was created.

Editing signals **3**

WHY THIS RESULT

Reasons for the decision

File: digital_invoice_edited.pdf

- This is a digital PDF: its pages were produced by software, not scanned or photographed, so there are no scanning or pixel-level editing traces for the image model to analyse. The pixel-based tampering score is therefore not used for this file.
- Created with: Microsoft Word, iLovePDF.
- The PDF was modified after it was created (created 28 Sep 2026 10:00, modified 29 Sep 2026 18:30).
- The PDF file was saved 2 times (incremental updates). This happens when an existing PDF is changed and saved again.
- The PDF was processed with a PDF editing tool (ilovepdf).
- Signs of editing were found in the file's metadata; compare the document with the original from its issuer.

PAGE 1 (RENDERED)

INVOICE INV-2026-0042
Date: 28/09/2026
Laptop x 1 45,000.00 35,000.00
Total 45,000.00 35,000.00

HOW IT WAS DECIDED

Analysis steps

1. The file type and structure were checked (Word, PowerPoint or PDF with a text layer).
2. The metadata was read: creating program, author, last editor, creation and modification dates,

Conclusion and Future Scope:

FinDoc AI provides a complete, explainable pipeline: any image, PDF, Word or PowerPoint file can be uploaded, scanned documents are analysed window by window with EfficientNetV2B0, and every result is explained with Grad-CAM, flagged regions and reasons. A careful document-level evaluation showed that detecting careful manual edits in scanned receipts is still hard: the honest test AUC is 47.04%, so the system should be used as a screening aid, not as proof.

- **Data:** More tampered training data, e.g. automatically generated forgeries
- **Text checks:** OCR-based checks that item prices, totals and tax add up
- **Models:** Forensic networks trained with pixel-level masks

THANK YOU